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Mountain Stream Ecosystem Metabolism and Nitrogen Cycling Responses to Hydroclimatic Volatility

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Abstract Changing hydroclimatic conditions increase streamflow variability and add uncertainty to predictions of watershed nutrient export and ecosystem function. Mountain headwater streams are especially sensitive to climate shifts due to their dependence on seasonal snowpack. We explored how changes in mountain precipitation affect the synchrony between hydrologic processes (peak flow and baseflow onset) and biological processes (metabolism and nutrient uptake) that control nitrogen dynamics. We monitored two mountain streams in the Lake Tahoe basin (California and Nevada, USA)—a larger open-canopy stream and a smaller, shaded stream—over two dry water years and one substantially wetter water year. On average, the large stream received more annual light, was warmer, and had higher inorganic nitrogen demand compared to the small stream. From dry to wet years, the onset of baseflow was delayed by ~40 days in both streams, shortening the optimal window for benthic algae growth by over a month and substantially reducing the potential for in stream nitrogen uptake. Concurrently, cumulative annual gross primary production declined by 86% (Δ −248 g O_2 m^{-2} yr^{-1}) in the large stream and 93% (Δ −26 g O_2 m^{-2} yr^{-1}) in the small stream. Mean daily ammonium supply increased by 47% (Δ + 0.01 g m^{-2} d^{-1}) in the large stream and by 112% (Δ + 0.16 g m^{-2} d^{-1}) in the small stream. These findings highlight the critical role of streams in regulating nutrient dynamics and show that even neighboring or hydrologically similar streams can exhibit distinct biogeochemical responses to hydroclimatic variability.

Plain Language Summary We investigated how the transition between anomalously dry and wet years altered the synchrony of flow and biological processes (e.g., nutrient uptake and ecosystem metabolism) that control nitrogen dynamics in two mountain streams. By comparing streams with different watershed sizes, runoff patterns, and precipitation accumulation within the Lake Tahoe basin (Nevada/California USA), we found that the larger less shaded stream supported higher metabolism fluxes and exhibited more efficient inorganic nitrogen (e.g., nitrate and ammonium) assimilation. However, nitrogen assimilation and retention were severely reduced during a transition from dry to wet conditions due to the delayed onset of baseflow, which caused most available nitrogen in both streams to be exported downstream rather than transformed through microbial processes or assimilated into biomass. This work demonstrates the high degree of variability in stream ecosystem function within mountain basins and highlights how hydroclimate volatility can not only cause variability in streamflow but also induce stream-specific changes in metabolism and nutrient cycling due to stream characteristics.

Key Points:

- Following a dry-to-wet hydroclimatic transition, the onset of baseflow in a small and large mountain stream was delayed by roughly 40 days
- The delay in baseflow onset in the wet year reduced cumulative annual gross primary production by over 90% in both streams
- Nitrogen demand dropped 80% during the wet year in the large stream, while ammonium export increased by more than 100% in the small stream

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

Current conceptual models of nutrient cycling and transport in the Northern Hemisphere have been shaped by historically cold and consistent winters; however, increasingly frequent warm winter melt and runoff events now interact with nutrient-enriched landscapes to mobilize large pulses of terrestrial nutrients to downstream waters, with largely unquantified impacts on water quality (Coasta et al., 2022; Coble et al., 2019; Seybold et al., 2022; Winter et al., 2022). These changes are especially pronounced in high-latitude and high-elevation ecosystems, where rapid climate-driven melt of snowpacks, glaciers, and permafrost is altering hydrologic regimes (Pepin et al., 2022; Sidle & Gomi, 2024; Viviroli et al., 2011). As warming intensifies, complex hydrologic responses are emerging, including highly variable atmospheric river events and abrupt shifts between contrasting moisture states (Guan et al., 2012; Rhoades et al., 2020). These dynamics reflect a broader pattern of "hydroclimatic

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whiplash," characterized by rapid transitions between extreme dry and extreme wet conditions, which challenge assumptions of hydroclimatic stationarity and alter the timing, magnitude, and connectivity of hydrologic and biogeochemical fluxes (Ficklin et al., 2022; Swain et al., 2025). High-elevation headwater mountain streams are important sources of water, carbon, and nitrogen for downstream freshwater ecosystems (Alexander et al., 2007; Gomi et al., 2002) and can rapidly process or export large quantities of carbon and nitrogen (Hotchkiss et al., 2015; Kerins & Li, 2023; Roberts et al., 2007). Improving our understanding of how mountain stream ecosystem function will change with increasing hydroclimatic volatility is essential for improved predictions of changes in downstream water quality.

The capacity of streams to transform and transport carbon and biologically limiting nutrients (e.g., nitrogen) is temporally dynamic and can reflect catchment characteristics associated with hydrologic connectivity, geochemistry, watershed area, and land use (Bernhardt et al., 2017; Bernot et al., 2010; Hotchkiss et al., 2018; Wohl & Scott, 2017). Within streams, the degree to which a given reach retains or transports nitrogen downstream varies spatially and temporally as a function of reach morphology, stream discharge, and biotic demand. As nitrogen travels downstream, it spirals through dissolved, particulate, and biotic compartments, at rates regulated by biological (e.g., uptake) or physical processes (e.g., hydrologic loss) (Covino et al., 2010; Ensign & Doyle, 2006; Li et al., 2021; Newbold et al., 1981). Whole ecosystem estimates of stream metabolism (gross primary productivity [GPP], ecosystem respiration [ER], and net ecosystem productivity [NEP]) indicate carbon consumption and fixation from both autotrophic and heterotrophic organisms across all instream habitats (Hoellein et al., 2013; Jankowski et al., 2021; Odum, 1956). Nutrient cycling within streams is closely related to metabolic activity as higher fluxes of GPP reflect the assimilatory demand of primary producers for nitrogen (Covino et al., 2018; Hall & Tank, 2003; Heffernan & Cohen, 2010).

Climate-driven shifts, including longer ice-free periods, elevated mean annual temperatures, and more variable flow regimes, are likely to create new seasonal windows for productivity and nutrient turnover (Bernhardt et al., 2018). Because stream metabolism and nitrogen cycling are tightly coupled to light and flow dynamics, earlier periods of low discharge and high light availability (e.g., prior to riparian leaf-out) in dry years may promote earlier algal growth, which can persist later into the fall and be sustained by low baseflows and nutrient-rich groundwater inputs (Bernhardt et al., 2022; Mulholland et al., 2006; Savoy et al., 2019; Vincent et al., 2025). In contrast, during wet years, increased streamflow, scour, and hydrologic connectivity may reduce biological retention, enhance physical transport, and mobilize stored nutrients from soils and streambeds, resulting in more downstream nutrient export and less biological nutrient uptake and assimilation (Mulholland et al., 2006; Seybold et al., 2022; Winters et al., 2022). In the Sierra Nevada (California and Nevada, USA), hydroclimatic variability has led to increased fluctuations in streamflow and nutrient export (Coats et al., 2008, 2016; Dettinger & Rajagopal, 2023; Domagalski et al., 2021; Sickman et al., 2002), which may reflect changes in the timing of key hydrologic events such as scour, peak flows, and baseflow onset, factors known to influence benthic algal persistence (Blaszczak et al., 2019; Francoeur & Biggs, 2006; Leathers et al., 2024; Uehlinger, 2000). Yet, how instream biotic communities regulate nitrogen cycling under these shifting conditions remains poorly understood, posing a critical gap in our ability to predict catchment-scale biogeochemical responses under future climate regimes marked by increasing hydroclimatic extremes (Swain et al., 2025).

In this study, we investigated how stream metabolism and nitrogen cycling co-varied along upper and lower reaches within two mountain streams draining contrasting catchments within the Sierra Nevada Lake Tahoe basin (California and Nevada, USA) in the northern Sierra Nevada mountains. These catchments differ in their cumulative annual precipitation and catchment morphologies, causing distinct flow regimes and water yields. We used the differences between sites to assess how streams transform, transport, and retain nitrogen across water years (2021–2024) that span a dry-to-wet "hydroclimatic whiplash event" in the US Pacific Southwest (Swain et al., 2025). We asked how do in-stream ecosystem metabolism and nitrogen retention and export vary (a) between contrasting catchments (one smaller and the other larger) and (b) among two dry (2021, 2022), one exceptionally wet (2023), and one average (2024) water years?

2. Materials and Methods

2.1. Study Area

To evaluate variation in stream productivity and nitrogen dynamics both within and among mountain streams, we monitored upper and lower stream reaches in both a large open canopy (Blackwood Creek) and small shaded

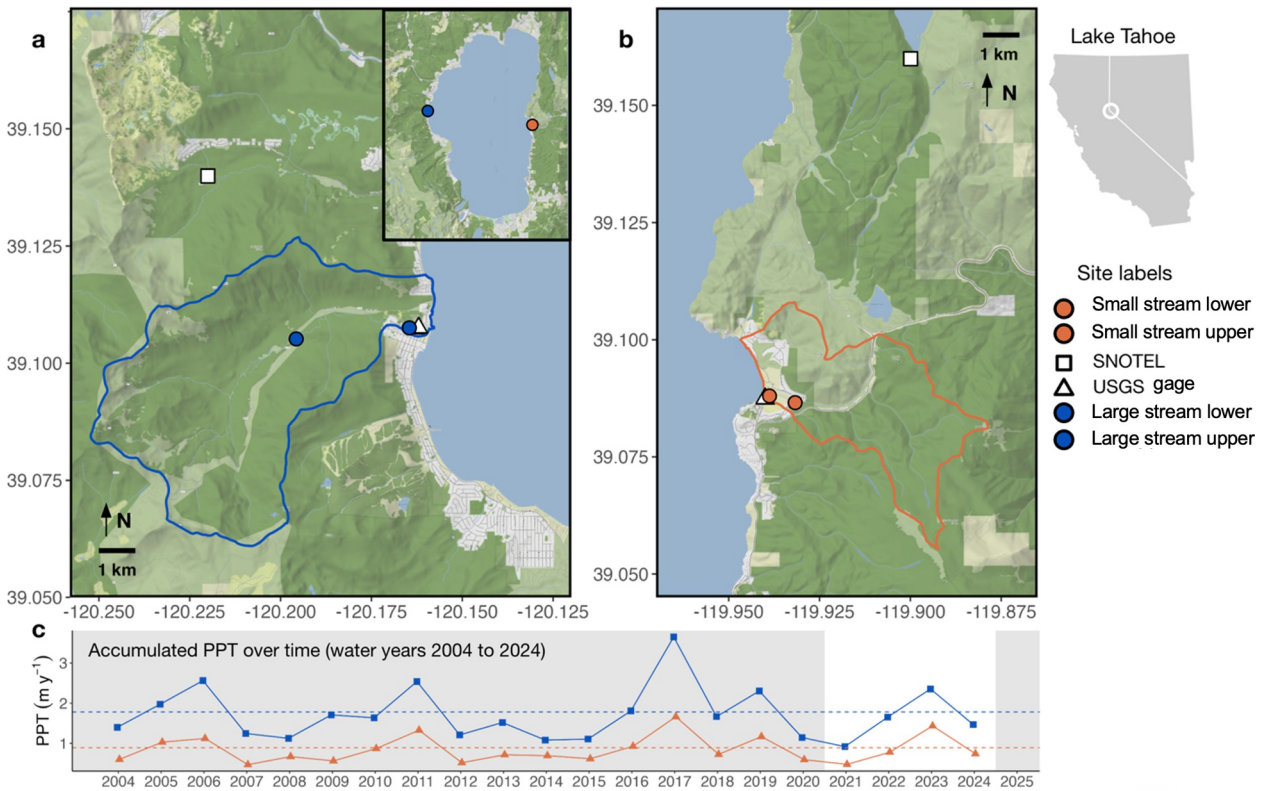


Figure 1. Map of the stream reach stations and nearby historical precipitation accumulation Blackwood creek, the larger stream in our study, is on the west side (a) and Glenbrook Creek (Small stream) on the east side of Lake Tahoe (b), located on the California-Nevada border. The drainage areas are outlined in dark blue for Blackwood and orange for Glenbrook Creek. Circles represent dissolved O_2 and specific conductance sensors at Blackwood (in blue), and at Glenbrook (in orange). Open black triangles show locations of United States Geological Survey stream gages and open black rectangles show the location of nearby SNOTEL stations. Annual accumulated precipitation (in m y^{-1}) from SNOTEL stations (squares) is shown during each water year (October 1st to September 30th) 2004–2024 (c). Color represents the stream association with BW in blue squares or GB in orange triangles. The dashed line represents the 20 years mean annual accumulated precipitation (BW: 1.72 m and GB: 0.85 m).

(Glenbrook Creek) mountain stream in the Lake Tahoe basin (California and Nevada, USA) from Spring 2021 to Fall 2024 (Figure 1). These two perennial streams from Lake Tahoe Basin sub-catchments represent the range of watershed sizes, modeled runoff, and precipitation accumulation within the basin (Dilts & Blaszczak, 2025) and have long-term 15-min discharge records maintained by the United States Geological Survey (USGS; Figure S1 in Supporting Information S1). We monitored paired upper and lower reaches in each stream to quantify short-scale longitudinal variation in metabolism and nutrient retention, as these processes remain poorly characterized in mountain streams. The large stream has mean ± standard deviation, annual daily flow of 0.95 ± 1.79 m^3 s^{-1} (USGS 2024b), a catchment area of 29.0 km^2 characterized by decomposed granite with notable volcanic and surficial deposits (Reuter & Miller, 2000) and is located on the western side of Lake Tahoe. The small stream is on the east shore of the lake, has a mean annual daily flow an order of magnitude less (0.0 ± 0.09 $m^3 s^{-1}$; USGS 2024a), and flows through a 10.7 km^2 watershed also characterized by decomposed granite. Both streams are pool–riffle systems with benthic substrates composed of mixed sand and cobble; however, the larger stream was characterized by more cobble dominance and a wider stream channel (mean width = 6.5 m), whereas the smaller stream had proportionally more sand cover and a narrower channel (mean width = 1.6 m). Both drainages are relatively undeveloped (<3% of total area; Dilts & Blaszczak, 2025), predominantly forested with conifers, but do contain nitrogen-fixing species such as alder (genus *Alnus*) capable of influencing stream nitrogen loads (Coats et al., 2016; Domagalski et al., 2021; Reuter & Miller, 2000).

2.2. Instream Instrumentation

We instrumented lower and upper reach stations within each creek with dissolved oxygen (O_2), water temperature, and conductivity sensors. We horizontally mounted sensors on cinder blocks positioned approximately

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0.1 m above the streambed in the thalweg, recording 15-min observations of dissolved O_2 (mg L^{-1}, miniDOTs Precision Management Engineering, Vista, California), specific conductance (SPC; in µS cm^{-1}, HOBO® U24 Conductivity Logger, Onset® Computer Corporation), and water temperature (°C, both sensors). Within the large stream, the lower reach station (39.11°N, 120.16°W) was installed from April 2021 to August 2024, immediately upstream of the USGS gage (500 m), while the upper reach station (39.105°N, 120.20°W) was established 4.15 km upstream of the lower station from June 2021 to October 2023. Within the small stream, both the lower reach station (39.09°N, 119.94°W; 200 m upstream of the USGS gage) and the upper reach station (39.09°N, 119.93°W; 1.10 km upstream of the lower station) collected data from March 2021 to October 2024. We cleaned all the sensors weekly in summer (June to September), bi-weekly in fall (September to November), and monthly in winter (December to May). Due to the potential for atmospheric river events to cause punctuated extreme flows, we occasionally removed sensors to avoid damage or loss of instrumentation (Table S1 in Supporting Information S1). All sensor data were filtered to remove observations affected by biofouling, maintenance, or sensor error and standardized to a common reference using post-deployment calibration offsets (Supporting Information S1 of Section 3.3).

2.3. Water Chemistry

We monitored water chemistry and collected sediment samples every week from May to August, every other week in September, and monthly from October to April at all stations except the upper reach in the large stream. The upper reach in the large stream was sampled less frequently because of limited winter and spring access. Instead, the upper reach was sampled weekly from July to August, every other week in September, and opportunistically in winter (Table S1 in Supporting Information S1). We collected and filtered duplicate water samples from randomized transects 10–50 m upstream of our instrumented reach stations (Section S1.1 in Supporting Information S1) using Whatman GF/F filters (0.7 µm pore size, Whatman, Piscataway, NJ, USA). Samples were kept cold immediately after collection and then frozen at −20°C for later chemistry analysis.

2.4. Sampling Stream Sediment

For estimates of sediment organic matter (OM), we collected surface sediment from transects 10–50 m upstream of our instrumented reach stations. We collected three subsamples of the top 5 cm of benthic sediments using a hand trowel (Section S1.1 in Supporting Information S1), combined and homogenized the three subsamples, sieved the composite sediment (2 mm), and took subsamples to measure sediment OM (in mg g^{-1}), sediment-water extractions, and sediment pH (Section S1.2 in Supporting Information S1).

2.5. Analyzing Surface Water and Sediment Chemistry

We analyzed all water and sediment chemistry at the University of Nevada, Reno (Section S1.2 in Supporting Information S1). We quantified sediment OM by weighing samples before and after drying them at 60°C for 48 hr and combusting them at 500°C for 8 hr. We measured sediment pH on a mixture of 3 g of wet sediment in 5 mL of 0.01 mol L^{-1} $CaCl_2$ (Orion Star A211 Benchtop pH Meter, Thermo Fisher Scientific, Waltham, Massachusetts, USA; Carter & Gregorich, 2007). We extracted sediment porewater by filtering the supernatant from a vortexed and centrifuged mixture of 3 ± 0.25 g of wet sediment and 25 mL of deionized water. We measured dissolved organic carbon (DOC) concentrations using a total organic carbon analyzer (TOC-V CPH; Shimadzu, Kyoto, Japan). We measured ammonium (NH_4^+- N with a detection limit of 0.002 mg N L^{-1}; US EPA, 1993a), orthophosphate (ortho-PO_4^{3-} with a detection limit of 0.402 µg P L^{-1}; US EPA, 1993c), and nitrate (NO_3^- - N with a detection limit 0.003 mg N L^{-1}; US EPA, 1993b) concentrations on a SEAL AQ400 discrete analyzer (SEAL Analytical, Mequon, Wisconsin, USA). Lastly, we set all sample chemistry values that were below the detection to half the minimum detection limit.

2.6. Environmental Covariates

We evaluated the spatiotemporal differences in light, precipitation, and stream flow across the large and small streams from dry (2021, 2022) to wet water years (2023; the water year is defined by the starting year of the 12-month period from October 1 to September 30 of the following year). We calculated the amount of light reaching the stream surface given canopy cover, annual precipitation accumulation in the catchment, and reach-specific streamflow dynamics. We estimated the amount of light at the stream surface (PAR in µmol m^{-2} s^{-1})

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using the biophysically based StreamLight model developed by Savoy et al., 2021; Section S2.1 in Supporting Information S1). We used precipitation (mm) data from two nearby Snow Telemetry stations (station 848, 39.14°N–120.22°W, at 2056 m ASL and station 615, 39.16°N–119.9°W, at 2403 m ASL) to estimate long-term (water years 2004–2024) precipitation regimes and contextualize the precipitation patterns observed in our study period. We obtained 15-min observations of streamflow and water stage from USGS stations for the large stream (gage: 10,336,660) and the small stream (gage: 10,336,730) and downloaded the data using the dataRetrieval package in R (De Cicco et al., 2026). Because USGS gage measurements represent site-scale conditions, we adjusted gage discharge and stage to derive reach-specific estimates of flow using empirical relationships based on local channel morphology (Section S3.1 in Supporting Information S1). We used the mean August discharge in each water year as a proxy for baseflow conditions to estimate the timing of baseflow onset.

2.7. Reach-Scale Stream Metabolism

We estimated daily GPP (g O_2 m^{-2} d^{-1}), ER (g O_2 m^{-2} d^{-1}), and temperature corrected gas exchange rates (K_{600}, d^{-1}) using the single-station (Holtgrieve et al., 2010; Odum, 1956) Bayesian modeling framework for the diel dissolved O_2 change method in the R package streamMetabolizer (Appling et al., 2018). We modeled metabolism at each reach from high frequency measurements of dissolved O_2 (mg L^{-1}) along with estimated surface light (PAR in µmol m^{-2} s^{-1}), reach specific mean daily flow (Q in m^3 s^{-1}), and mean reach depth (m). Equilibrium O_2 saturation concentration (mg L^{-1}) was calculated using water temperature and barometric pressure in the Garcia-Benson equation, and the observed dissolved O_2 was subtracted from the estimated equilibrium O_2 to calculate the O_2 saturation deficit (Section S3.3 in Supporting Information S1).

We used different approaches for parameterizing gas exchange in the large and small stream metabolism models (Section S3.4 in Supporting Information S1). For the large stream, K_{600} was estimated within the metabolism model using discharge-binned priors to account for variation in streamflow and allow the priors for K_{600} to be shared among days with similar flows (five bins, Table S3 in Supporting Information S1). K_{600} priors for each bin were informed by the mean gas exchange rates measured in the lower reach using the inert gas tracer method for argon to nitrogen ratios across three seasonal flows (Hall & Madinger, 2018; Section S3.2 in Supporting Information S1). This approach allowed daily K_{600} to be estimated as a model parameter within the upper and lower reach while setting prior constraints to account for reach-specific geomorphology related to gas-exchange. For the small stream, we used a fixed log-linear relationship between measured K_{600} and streamflow. The intercept and slope of this relationship were estimated from gas exchange measurements across four season flows in the lower reach and incorporated into the metabolism model as highly constrained priors on the regression coefficients (Hall & Madinger, 2018; Section S3.2 and S3.4 in Supporting Information S1). We additionally only modeled metabolism on days where streamflow was within the range represented by our gas exchange measurements to avoid extrapolating modeled gas exchange and flow relationships beyond our observed conditions.

We ran each Markov Chain Monte Carlo algorithm in stream Metabolizer with four chains, 5,000 iterations total (2,500 iterations of burn in), a target acceptance rate (adapt delta) of 0.98, a maximum tree depth of 18, and a thinning interval of 1 (Section S3.5 in Supporting Information S1).

We evaluated metabolism estimates and model performance to assess model fit and how well models produced ecologically realistic values. We assessed chain convergence using the Gelman-Rubin statistic ($\hat{R}$) for each parameter in the model (GPP, ER, and K_{600}), removing days where $\hat{R} > 1.05$ for any estimated parameter. We assessed the relationship between ER and K_{600} (Pearson correlation coefficient via the cor function in base R) and calculated normalized root-mean-squared errors (NRMSE) for daily model fits of predicted and observed dissolved O_2 values and we removed estimates on days with high error (NRMSE > 0.3). We removed days where metabolism models produced biologically unrealistic estimates (95% credible interval for daily GPP was <0 and days when the lower 95% credible interval for daily ER was >0, Table S5 in Supporting Information S1).

2.8. Reach-Scale Nutrient Supply and Demand

We calculated nitrogen supply for reaches (as either NO_3^- - N or NH_4^+- N in g N m^{-2} d $^{-1}$) based on stream discharge (Q; in L s^{-1}), nitrogen concentrations (C; in g N L^{-1}), stream width (w; in m), and reach length (L; in m) (Covino et al., 2018).

$$\text{Nitrogen supply} = \frac{86400 \times Q \times C}{w \times L} \quad (1)$$

We used daily estimates of GPP and ER to calculate the in-stream demand for nitrogen (g N m^{-2} d^{-1}), defined as metabolism-based assimilation from autotrophic and heterotrophic processes (Covino et al., 2018; Section S4 in Supporting Information S1). We used a photosynthetic coefficient of 1, an autotrophic respiration coefficient (r_a) of 0.5 (r_{aGPP}; Hall & Tank, 2003; Covino et al., 2018), and an autotrophic M C:N of 16:1 recommended for low-nitrogen streams (Stelzer & Lamberti, 2001) to estimate autotrophic respiration. We estimated heterotrophic nitrogen assimilation using the low energy coefficient for heterotrophic growth efficiency (HGE = 0.05) and a heterotrophic molar ratio of C:N of 20:1 (Hall & Tank, 2003). To understand the proportion of nitrogen conveyed between stream reaches, we estimated the amount of nitrogen retained between upper and lower stream reach stations (in g N m^{-2} d^{-1}) as:

$$\text{Nitrogen retention} = 1 - \frac{\text{Nitrogen supply upper station}}{\text{Nitrogen supply lower station}} \quad (2)$$

Lastly, we used the weighted regressions on time, discharge, and season (WRTDS) model in the EGRET package (Hirsch et al., 2024) to infill additional observations of NO_3^--N (upper reach large stream = 4 samples, lower reach small stream = 8 samples, and upper reach small stream = 11 samples) or NH_4^+-N at each reach (lower reach large stream = 22 samples, upper reach large stream = 5 samples, lower reach small stream = 10 samples, upper reach small stream = 12 samples) between discrete sampling dates, increase the temporal resolution of nutrient concentration data across each reach (SI Section 1.3; Figure S2 in Supporting Information S1).

2.9. Statistical Analyses

To assess spatial and temporal differences in metabolism and nitrogen dynamics, we used analysis of variance to test for variation in nitrogen supply and demand among reach locations (the upper or lower reaches of the large or small stream) and among water years (dry or wet). We reported the percent change, F-statistics, and p-values for results with statistically significant differences ($p < 0.05$).

To evaluate potential controls on stream metabolism, we fit separate mixed-effects models for the lower reach of the large stream and the upper reach of the small stream as these locations contained the most continuous metabolism observations. We evaluated relationships between GPP and mean daily light at the stream surface and mean daily streamflow, and between |ER| and mean daily streamflow and mean daily water temperature. We implemented models with a first-order autoregressive correlation (ACF) structure to account for temporal dependence (Pinheiro, 2011; Zuur et al., 2009). All predictor variables were centered and scaled (mean = 0, standard deviation = 1) prior to model fitting.

We conducted all analyses in R version 4.4.2 (R Core Team, 2024). We fit GLMMs using the lmer package based on maximum-likelihood based estimation (Bolker et al., 2009). For all GLMMs, we assessed the significance ($p < 0.05$) of covariates, evaluated residual normality, and calculated the model R^2 using rsquaredglmm (Barton & Barton, 2015).

3. Results

3.1. Hydroclimate Differences Across Catchments From Dry to Wet Water Years

Our observation period coincided with a cycle of hydrologic extremes and the higher precipitation and streamflow during the wet water year (2023), which resulted in increases in peak flow magnitude and delayed the onset of baseflow in both streams. Water year 2023 was one of the top five wettest years in the Sierra Nevada mountains for the period of record (1896–2025), the preceding two water years (2021 and 2022) were impacted by prolonged drought, and 2024 was near average for snowpack accumulation (California Department of Water Resources, 2024). Relative to the wet year, mean annual precipitation in the preceding dry years was 45% lower in the large catchment and 56% lower in the small catchment (Figure 1c), while annual water yield was 65% and 87% lower. During the wet year relative to the dry years, the peak of specific discharge was 65% higher at the large stream (Δ signifies magnitude change from dry to wet water years, Δ +0.30 m^3 s^{-1} km^{-2}) and 91% higher at the

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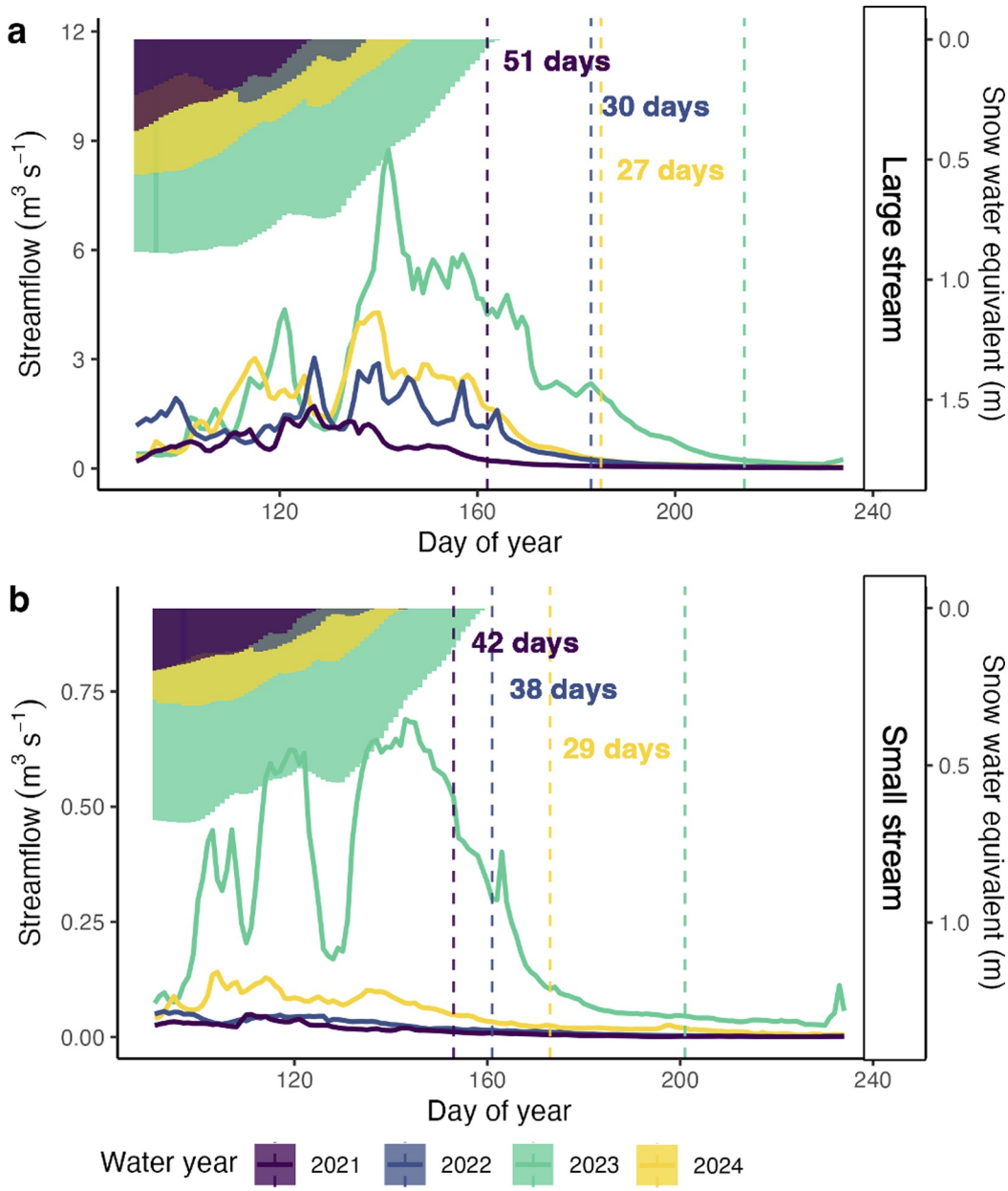


Figure 2. Daily streamflow (m^3 s^{-1}; solid lines) and nearby accumulated snow water equivalence (m; hanging filled areas) at the United States Geological Survey gages at the outlets of the large creek (a) and the small creek (b) from April 1st to September 1st (day of year 91 to day of year 244) for water years 2021 (purple), 2022 (blue), 2023 (teal), and 2024 (yellow). The vertical line represents the annual onset of base flow (calculated as the mean streamflow in August each year) for each water year. The text in the top right corner of each panel represents the difference in days from the onset of baseflow in the wet year (2023) to every other water year.

small stream (Δ +0.07 m^3 s^{-1} km^{-2}). The mean onset of base flow occurred more than 40 days later for both creeks in the wet year relative to the dry water years, causing a 32% decrease in the length of the optimal growing season (days since the onset of baseflow to October 31st) at the large stream (Δ − 42 days) and a 30% decrease in the length of the optimal growing season at the small stream (Δ −44 days; Figure 2).

Both streams exhibited substantial spatial and temporal heterogeneity under environmental conditions (Table 2). Wetter conditions led to colder water temperatures and lower sediment OM in the two streams. Surface light differed primarily between streams, with the large stream receiving 23% more mean annual light (266 µmol m^{-2} s^{-1}) than the small stream (204 µmol m^{-2} s^{-1}). Within the large stream, surface light availability

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Table 1
Water Chemistry in Sediment Water (Sediment) or Surface Water (Water) From the Large Stream and the Small Stream at the Upper and Lower Reach Locations

Stream	Reach	Type	NO_3^- mean (μg L^{-1})	NO_3^- range (μg L^{-1})	NH_4^+ mean (μg L^{-1})	NH_4^+ range (μg L^{-1})	Mean pH
large	lower	sediment	69.5	10.0–434.0	25.8	2.0–87.1	7.39
large	lower	water	12.6	1.5–57.0	4.6	2.0–11.9	7.52
large	upper	sediment	51.6	13.0–143.0	13.5	2.0–78.6	7.07
large	upper	water	12.6	1.5–57.0	8.4	2.0–20.8	7.09
small	lower	sediment	104.0	9.0–778.0	44.2	2.4–177.4	7.76
small	lower	water	17.7	1.5–47.0	7.6	2.0–17.1	7.67
small	upper	sediment	63.1	8.0–482.0	24.0	2.0–128.8	8.10
small	upper	water	18.5	1.5–51.0	8.5	2.0–22.0	8.07

Note. Analyses include nitrate (NO_3^- - N in μg L^{-1}), ammonium (NH_4^+- N in μg L^{-1}), and pH. Surface water summaries include values infilled from Weighted Regressions on Time, Discharge, and Season model (Section S1.3 in Supporting Information S1).

was 38% higher in the lower reach (Δ +89 μmol m^{-2} s^{-1}), whereas in the small stream, downstream light availability was only 10% higher relative to the upper reach (Δ +22 μmol m^{-2} s^{-1}). Interannual differences in modeled surface light may have reflected changes in riparian canopy conditions due to heavy snowfall or delayed leaf emergence; but we did not measure these potential mechanisms in this study. The mean annual water temperature was higher in the large stream (11.4°C) than in the small stream (8.5°C; Table 2). Interannually, mean water temperature declined from dry to wet years in both streams but more strongly in the large stream (up to Δ −3.2°C, 27% at the lower reach of the large stream; Table 2). Sediment OM concentrations were on average 75% lower in the large stream (16.6 mg g^{-1}) relative to the small stream (22.2 mg g^{-1}). Within streams, sediment OM decreased by 15% downstream in the large stream (Δ −1.5 mg g^{-1}) but increased by 24% downstream in the small stream (Δ +4.9 mg g^{-1}). From dry to wet years sediment OM declined modestly in the large stream (lower reach: 4%–11%) but substantially in the small stream (lower reach: 38%, Δ −12.9 mg g^{-1}; upper reach: 42%, Δ −12.5 mg g^{-1}, Table 2, Figure S14 in Supporting Information S1).

3.2. Stream Metabolism

The proportion of days meeting metabolism quality-screening criteria varied considerably among reaches, and therefore subsequent analyses of environmental controls, temporal metabolism patterns, and nitrogen demand were restricted to daily metabolism estimates that met our model convergence and plausibility criteria (Section S3.6 and Table S5 in Supporting Information S1). We retained 57% of days (n = 493) in the lower reach of the large stream, 67% of days (n = 290) in the upper reach of the large stream, 40% of days (n = 408) in the lower reach of the small stream, and 33% of days (n = 335) in the upper reach of the small stream (Tables S1 and S5 in Supporting Information S1) out of the 4 years of nearly continuous observations to reflect reasonable estimates of metabolism (Section S3.3 and Table S5 in Supporting Information S1). Despite incorporating empirical estimates of gas exchange during model fitting, we observed a positive Pearson's correlation between ER and K_{600} in the upper reach of the large stream (r = 0.66), no significant correlation between ER and K_{600} at either the lower reach of the large stream (r = −0.04) or the upper reach of the small stream (r = −0.11), and a negative correlation in the lower reach of the small stream (r = −0.66, Figures S9–S12 in Supporting Information S1). Relationships between ER and K_{600} varied across reaches, indicating that correlations between metabolism and gas exchange parameters were not consistent across sites. The positive ER and K_{600} relationship in the upper reach of the large stream (r = 0.66) likely reflects hydrologic controls on both gas exchange and biological activity, where lower-flow conditions corresponded with reduced gas exchange and higher rates of GPP and ER. In contrast, the negative ER and K_{600} relationships suggest reduced identifiability of ER at these reaches, particularly during periods with weak biological signals. As a result, extreme daily metabolism estimates are likely less reliable than the overall temporal patterns and central tendencies.

Negative NEP indicated heterotrophic conditions across all reaches (NEP range: −22.6 to −0.6 g O_2 m^{-2} d^{-1}) with the highest fluxes of GPP and ER occurring within the large stream (Figure 2). Including both upper and lower reaches, both GPP and ER magnitudes were higher in the large stream than in the small stream. Mean GPP

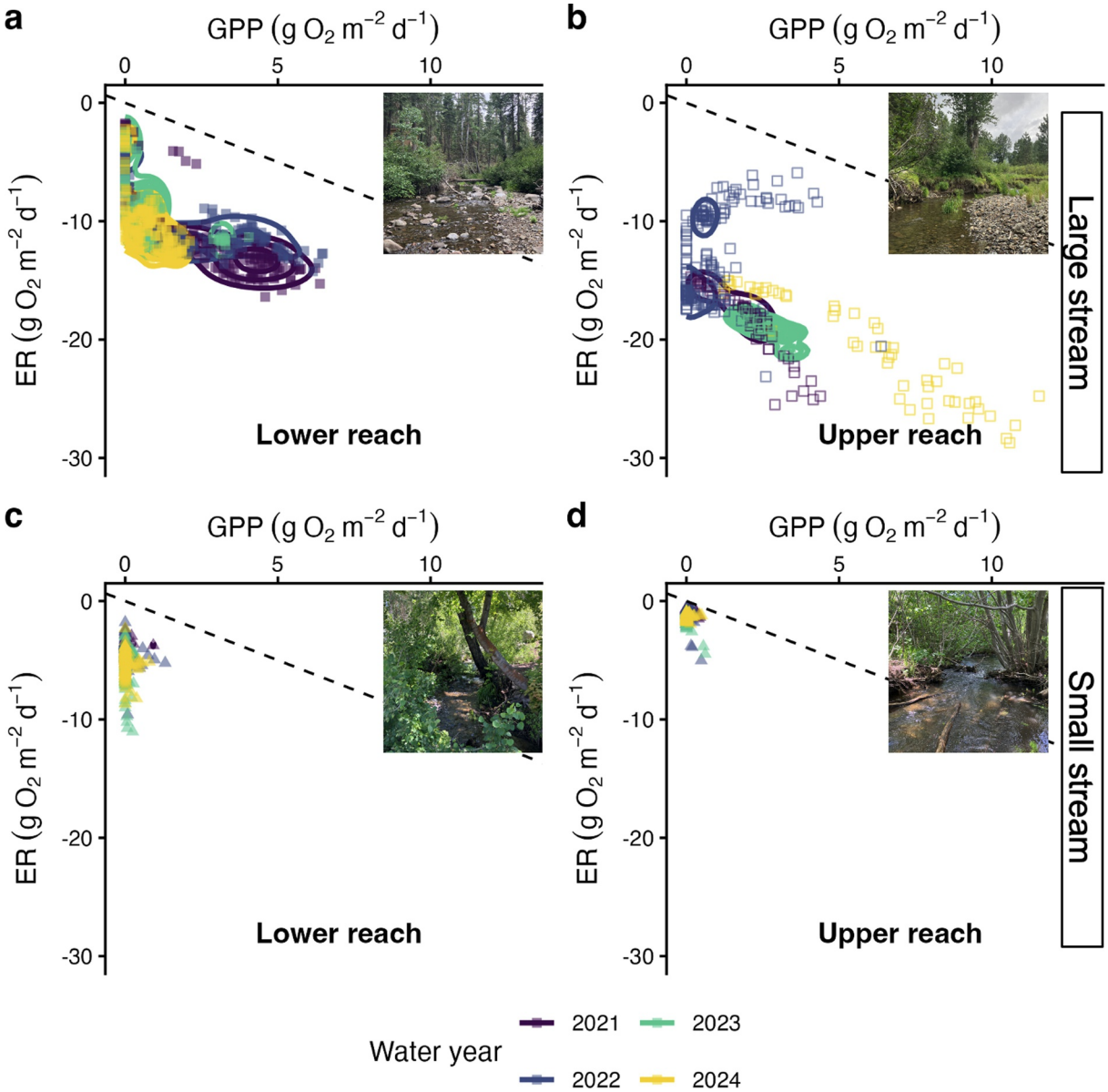


Figure 3. Daily gross primary productivity (GPP) and ecosystem respiration at four reach stations (a) the lower (n = 493 days, filled squares), (b) and upper (n = 290 days, open squares) reach locations in the large creek, and the (c) lower (n = 408 days, filled triangles), and (d) upper (n = 335 days, open triangles) reach locations in the small creek. Color represents the water year 2021 (purple), 2022 (blue), 2023 (teal), and 2024 (yellow), and the 1:1 line is in black, where values below represent heterotrophic days.

was 1.9 g O_2 m^{-2} d^{-1} compared to 0.05 g O_2 m^{-2} d^{-1} in the small stream, whereas mean |ER| in the large stream was 12.3 g O_2 m^{-2} d^{-1} and 3.2 g O_2 m^{-2} d^{-1} in the small stream (Figure 3).

3.3. Stream Metabolism in Relationship to Environmental Conditions

Comparing results from the large and small streams, we found contrasting relationships between GPP and light as well as |ER| and water temperature. GPP and |ER| were positively correlated (Pearson correlation coefficients) at both reaches in the large stream ($r > 0.5$; Figures S9 and S10 in Supporting Information S1), while |ER| and GPP were weakly correlated in the small stream ($r < 0.15$; Figures S11 and S12 in Supporting Information S1).

After accounting for temporal autocorrelation (φ range: 0.72 to 0.99; Figure S16 in Supporting Information S1), metabolism responses differed between streams (Section S5.1 in Supporting Information S1). In the large stream,

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Table 2
Mean Environmental and Physicochemical Conditions Measured at Upper and Lower Reaches of Large and Small Streams Across Water Years 2021–2023

Stream	Reach	Water year	Flow ($m^3 s^{-1}$	Temperature (°C)	SPC (μS cm^{-1})	LAI	PAR (μmol m^{-2} s^{-1})	OM (mg g^{-1})
large	lower	2021	0.326	13.93	60.3	1.97	246	15.4
large	lower	2022	0.635	9.34	57.2	1.61	220	17.0
large	lower	2023	1.159	8.49	36.3	1.51	237	15.6
large	upper	2021	0.019	12.58	37.3	2.76	145	12.6
large	upper	2022	0.142	12.78	43.1	2.05	138	23.8
large	upper	2023	0.188	10.98	35.5	1.78	152	16.2
small	lower	2021	0.011	8.55	324.6	1.79	216	31.3
small	lower	2022	0.016	8.01	329.2	1.41	181	38.9
small	lower	2023	0.133	6.94	239.5	1.33	196	22.2
small	upper	2021	0.016	9.78	360.5	1.59	209	30.9
small	upper	2022	0.022	9.37	365.2	1.26	162	29.2
small	upper	2023	0.179	8.12	261.3	1.22	172	17.5

Note. Variables include stream discharge (flow; $m^3 s^{-1}$), water temperature (°C), specific conductance (SPC; μS cm^{-1}), leaf area index (LAI), photosynthetically active radiation at the stream surface (PAR; μmol m^{-2} s^{-1}), and sediment organic matter content (OM; mg g^{-1}).

neither GPP nor |ER| were significantly related to the environmental predictors we examined ($p > 0.05$). In the small stream, GPP was positively associated with mean daily light (β_{light}: 0.02 ± 0.01, $p = 0.05$), whereas |ER| was negatively associated with streamflow (β_{flow}: −0.31 ± 0.06, $p < 0.001$).

3.4. Stream Metabolism Responses to a Dry-to-Wet Water Year Transition

GPP and |ER| declined substantially from dry-to-wet hydroclimatic conditions in all reaches, except the lower reach in the small stream, which exhibited a very small signal of metabolism throughout our study. Within the large stream, cumulative annual GPP decreased by 86% at the lower reach (Δ −298.4 g O_2 m^{-2} yr^{-1}, 144 days shared among years, Figure S15 in Supporting Information S1) and by 97% at the upper reach (Δ −197.2 g O_2 m^{-2} yr^{-1}, 58 days shared among years) from dry to wet water years. Cumulative |ER| in the large stream similarly decreased by 50% (Δ −611.1 g O_2 m^{-2} yr^{-1}, 144 days shared among years) at the lower reach and by 26% at the upper reach (Δ −264.2 g O_2 m^{-2} yr^{-1}, 58 days shared among years) from dry to wet years. In the small stream, cumulative GPP in the lower reach was low (>5 g O_2 m^{-2} yr^{-1}) and remained consistent from dry to wet water years, while cumulative GPP in the upper reach decreased by 99% (Δ −6.8 g O_2 m^{-2} yr^{-1}, 208 days shared among years). Cumulative |ER| increased by 53% (Δ + 57.0 g O_2 m^{-2} yr^{-1}, 113 days shared among years) at the lower reach and decreased by 27% (Δ −26.9 g O_2 m^{-2} yr^{-1}, 208 days shared among years) at the upper reach in the small stream from dry to wet water years (Table S7 in Supporting Information S1).

3.5. Nitrogen Concentrations, Supply, and Demand Within the Large and Small Stream

We observed consistently low nitrogen concentrations in surface waters and sediment porewater and a temporal decoupling between peak in-stream nitrogen demand and supply seasonally across both streams (Table 1). Most nutrient concentrations across all samples (97% NO_3^--N, $n = 251$; 83% of NH_4^+-N, $n = 180$; 100% of ortho-PO_4^{3-}, $n = 241$; and 91% of DOC $n = 162$) were above detection limits. However, all nutrient concentrations were generally low and the mean surface water DIN:P molar ratio across all stream sites was 14.2. The small stream had higher mean nitrate (18.1 μg L^{-1}) and ammonium (8.1 μg L^{-1}) concentrations relative to the large stream (NO_3^--N = 12.6 μg L^{-1}, NH_4^+-N = 6.6 μg L^{-1}; Table 1). In the large stream, nitrate concentrations tended to increase downstream while ammonium concentrations decreased downstream, and we observed the opposite pattern in the small stream.

Nitrogen supply (mean = 0.038 g m^{-2} d^{-1}) was consistently higher than nitrogen demand across all reaches, (mean supply - demand; NO_3^--N: Δ 0.274 g m^{-2} d^{-1} and NH_4^+ -N: Δ 0.043 g m^{-2} d^{-1}), and nitrogen demand only exceeded supply on 37% of days, primarily in the large stream during the fall (Figure 4a). Nitrogen supply and demand were temporally decoupled, especially in the large stream, where peak supply occurred in early

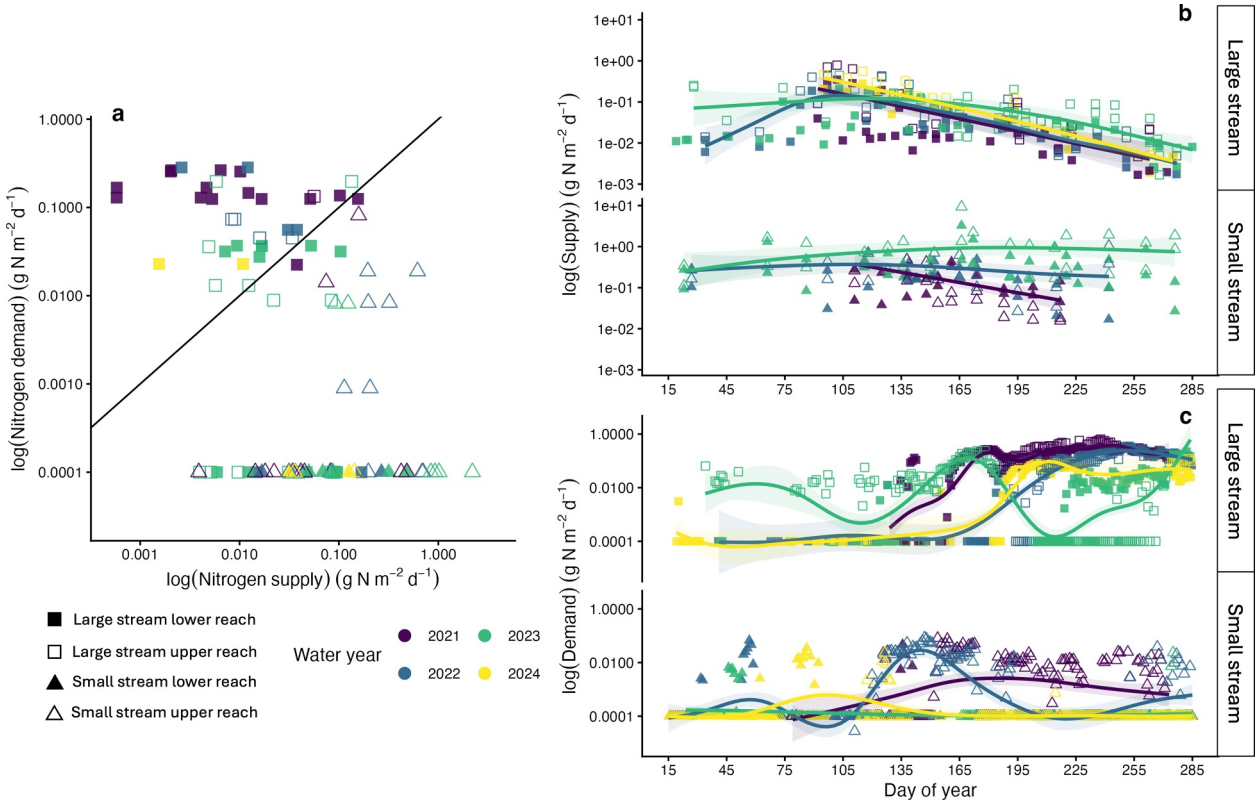


Figure 4. (a) Log transformed nitrogen demand (in g m^{-2} d^{-1}) and log-transformed nitrogen supply (in mg m^{-2} d^{-1}) for nitrate (NO_3^- - N) and ammonium (NH_4^+ - N). The black line represents the 1:1 line between demand and supply. (b) Log transformed nitrogen supply for NO_3^- - N and NH_4^+ - N on each day of year (15–274 as January 1^{5th} to October 1st) for large (upper panel) or small creek (lower panel). (c) Log transformed nitrogen demand for the day of year for large (upper panel) and small (lower panel). Point color represents water year 2021 (purple), 2022 (blue), 2023 (teal), and 2024 (yellow) and point shape represents reach station with the lower (filled squares), and upper (outlined squares) reaches in the large creek, as well as the lower (filled triangles), and upper (outlined triangles) reach locations in the small creek. The lines in panels a and b are a generalized additive model with a smoothing spline ($k = 10$), colored by water year, with shading to represent the 95% confidence interval around the fitted line.

summer, but demand peaked weeks later in late summer. In contrast, nitrogen supply in the small stream was more evenly distributed, with a spring peak and negligible demand throughout the year (Figures 4b and 4c).

3.6. Nitrogen Dynamics in the Large and Small Stream Across dry and Wet Years

From dry to wet years, nitrogen supply increased across both streams, while nitrogen demand declined sharply (Figure 5). In the large stream, the mean daily nitrate supply was unchanged but the mean daily ammonium supply increased by 47% (lower reach: Δ +0.007 g m^{-2} d^{-1}, $F_{2,54}$: 3.3, $p = 0.04$, and upper reach: Δ +0.024 g m^{-2} d^{-1}, $p > 0.05$). These increases in supply coincided with an 80% decrease in mean daily nitrogen demand (lower reach: Δ −0.121 g m^{-2} d^{-1}, $F_{2,354}$: 76.2, $p < 0.01$ and upper reach: Δ −0.154 g m^{-2} d^{-1}, $F_{2,263}$: 166.2, $p < 0.01$). In the small stream, mean daily nitrate supply (lower reach: Δ + 0.437 g m^{-2} d^{-1}, $F_{2,40}$: 4.11, $p = 0.02$, and upper reach: Δ +1.279 g m^{-2} d^{-1}, $F_{2,36}$: 3.95, $p = 0.03$) and mean daily ammonium supply increased substantially at both reaches (lower reach: Δ +0.150 g m^{-2} d^{-1}, $F_{2,40}$: 5.89, $p < 0.01$ and upper reach: Δ +0.364 g m^{-2} d^{-1}, $F_{2,36}$: 3.47, $p = 0.04$), while average daily nitrogen demand, remained low at the lower reach and declined by 94% at the upper reach station (Δ −0.001 g m^{-2} d^{-1}, $F_{2,423}$: 7.89, $p < 0.01$).

Throughout the entire survey period, the large stream retained a more inorganic nitrogen relative to the small stream. Positive estimates of nitrogen retention between the upper and lower reaches in the large stream suggested substantial downstream removal (NO_3^--N: mean = 0.44 g m^{-2} d^{-1}; range = −0.16 to 0.87 g m^{-2} d^{-1}; NH_4^+-N: 0.04 g m^{-2} d^{-1}; −3.91 to 0.94 g m^{-2} d^{-1}). Negative retention estimates in the small stream (NO_3^--N: −0.71 g m^{-2} d^{-1}; −8.70 to 0.95 g m^{-2} d^{-1}; NH_4^+-N: −0.72 g m^{-2} d^{-1}; −8.16 to 0.99 g m^{-2} d^{-1}) suggest external inputs of inorganic nitrogen occurred between sampling reaches.

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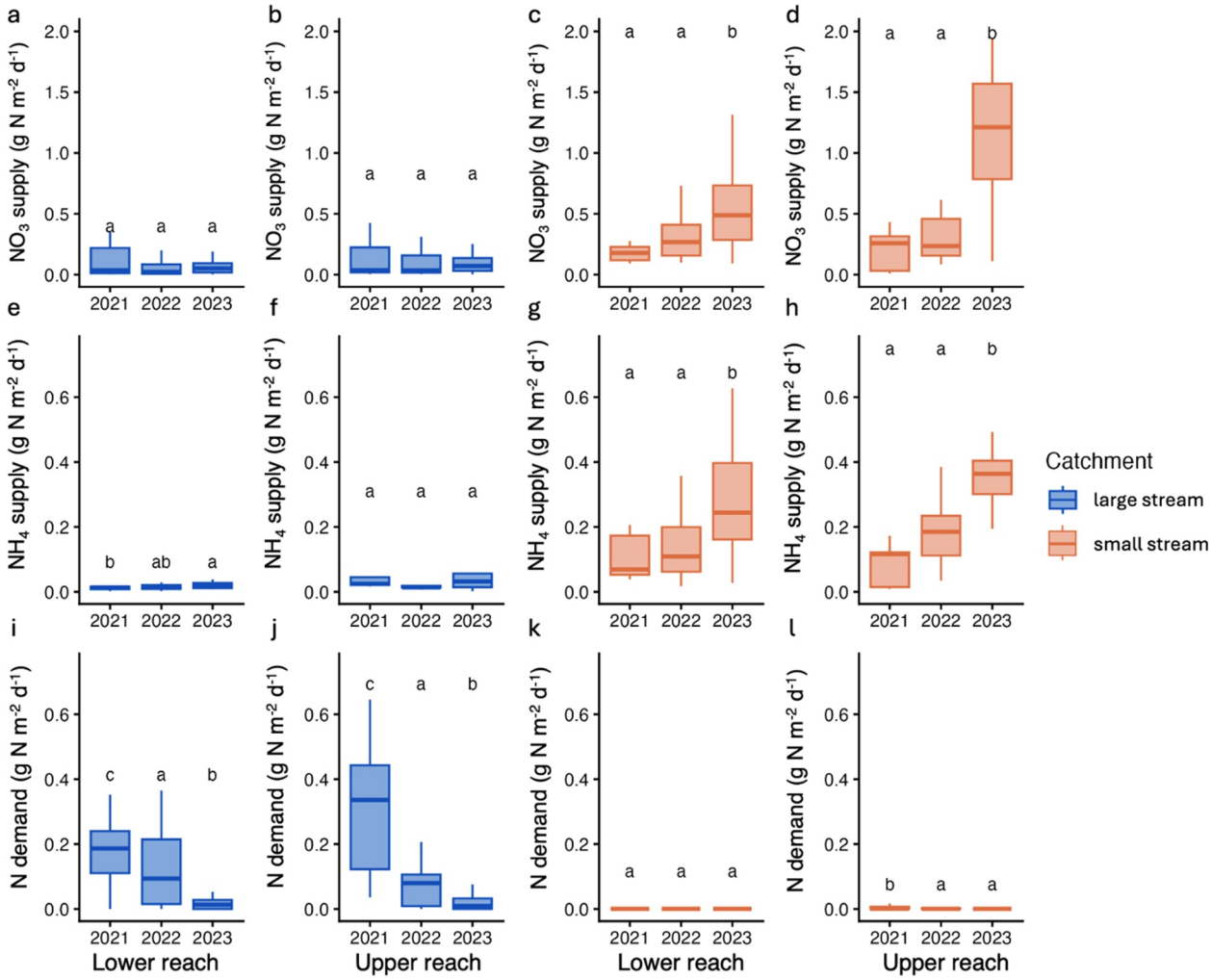


Figure 5. Mean annual estimates of nitrogen dynamics at lower (far left), and upper (middle left) reach locations in the large creek (blue), as well as the lower (middle right), and upper (far right) of the small creek (orange) from dry (2021 and 2022) to wet (2023) water years. a–d represent mean annual nitrate supply (as NO_3^- - N in g m^{-2} d^{-1}). e–h represent mean annual ammonium supply (as NH_4^+ - N in g m^{-2} d^{-1}). i–l represent mean annual inorganic nitrogen demand (n g m^{-2} d^{-1}). Letters above each box correspond to Tukey's Honestly Significant Differences that represent statistical differences ($p < 0.05$) among years using ANOVA.

The proportion of nitrate and ammonium retained between stream reach stations decreased (NO_3^--N: by 42% Δ −0.20 g N L^{-1} d^{-1} and NH_4^+-N: 150% Δ −0.23 g m^{-2} d^{-1}; $F_{2,\ 10}$: 38.3, $p = 0.007$) in the large stream from dry to wet conditions, indicating a switch from nitrogen processing to export. In the small stream, between-reach retention of nitrate (−1.22 to −0.14 g N L L^{-1} d^{-1}) and ammonium increased (−1.21 to −0.09 g m^{-2} d^{-1}); however, the negative values are indicative of additional inorganic nitrogen sources and suggest that more nitrogen was exported relative to be being retained during the wet water year.

4. Discussion

In this study, we examined how stream ecosystem metabolism and nitrogen cycling, responded longitudinally to a rapid dry-to-wet hydroclimatic transition. By capturing the water years that spanned a dry-to-wet "hydroclimatic whiplash" event (Swain et al., 2025), we evaluated how a stream draining a smaller, steeper, and drier catchment (Glenbrook Creek) and a larger, flatter, and wetter catchment (Blackwood Creek), responded to these hydroclimatic extremes. Our results demonstrate that streams differ in their capacity to assimilate versus transport inorganic nitrogen across these contrasting hydroclimatic and catchment conditions. The large stream consistently exhibited higher metabolic fluxes across both upper and lower reaches and under both dry and wet hydrologic conditions. Despite experiencing more frequent scouring flows and greater intra-annual variability, the

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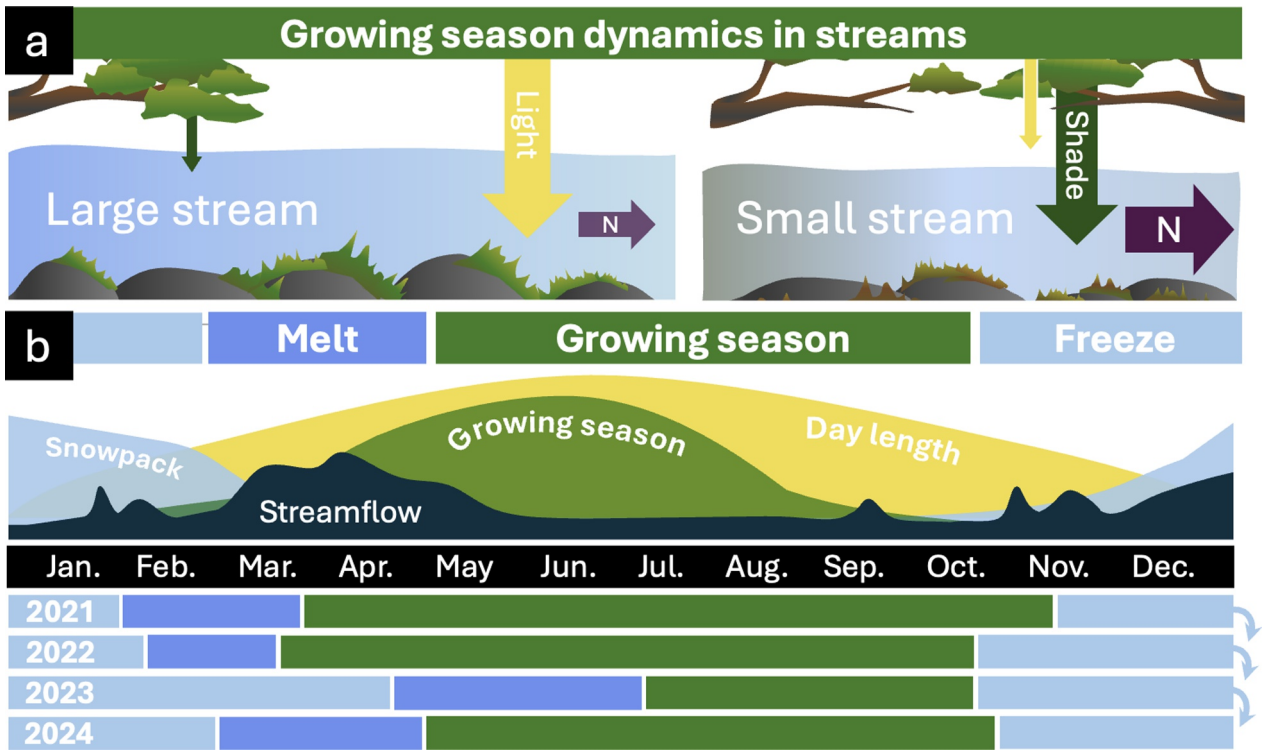


Figure 6. Hydroclimatic variability influenced growing-season length, ecosystem metabolism, and nitrogen cycling in the large and the small mountain streams. Panel (a) illustrates how differences in light availability, channel morphology, and hydrologic conditions contributed to contrasting patterns of primary production, organic matter accumulation, and nitrogen processing between the two streams. Panel (b) illustrates the expected annual timing of the growing season in mountain streams and how hydroclimatic variability can shift its timing by altering snowpack duration and the onset of snowmelt and peak flow. During the wet year (2023), delayed snowmelt postponed the onset of baseflow.

large stream's greater width, depth, and light availability likely supported more periphyton growth and higher GPP, compared to the smaller and more shaded stream (Figures 3 and 6). These differences in GPP were closely associated with differences in nitrogen demand and light availability at the stream surface. In the large stream, nitrate and ammonium supply closely matched modeled nitrogen demand, suggesting a higher proportion of biological assimilation (Figure 4; Covino et al., 2018). In contrast, the small stream exhibited lower GPP and ER, and nitrogen demand was well below supply, indicating weaker biological demand and greater downstream export (Figure 4). These differences in nitrogen assimilation potential led to a high export of inorganic nitrogen, particularly ammonium, from the small stream during the wet water year (Figure 5), with implications for downstream nutrient loading and water quality in an oligotrophic lake.

4.1. Hydrologic Conditions Expand or Contract Windows of Ecosystem Productivity and Nitrogen Retention

Antecedent flow conditions influenced stream metabolism in our study by delaying the onset of baseflow, and thereby the optimal window for stream productivity by over a month in the wet water year. In mountain streams, variation in precipitation accumulation and run-off ratios determine stream discharge (Hammond et al., 2024; Poff & Ward, 1989), which act as physical filters to when and where in-stream communities can persist (Uehlinger & Naegeli, 1998). Cycles of drought punctuated by intense precipitation are common in the western U.S., reflecting large-scale climatic oscillations that are expected to intensify in the future (Cole et al., 2002; Guan et al., 2012; Swain et al., 2025). Our study captured a cycle of drought and wetness, where dry conditions led to larger (2021 and 2022) or smaller (2023) windows of opportunity (O'Donnell & Hotchkiss, 2022; Uehlinger et al., 2010) for algal growth depending on the timing of peak flow recession and baseflow onset (Figure 2). In the wet year (2023), the onset of baseflow occurred approximately 40 days later than in dry years, resulting in a 30%–32% reduction in the optimal growing season length across both streams. This shorter growing season corresponded with substantial declines in seasonal GPP, with cumulative annual GPP reduced by 74%–98% and

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|ER| reduced by 26%–97% across our stream reaches (Figure 2; Figure S15 and Table S7 in Supporting Information S1).

While reductions in |ER| were correlated with reductions in GPP in our study streams, higher flows can also increase |ER|, resulting in lower NEP (Roberts & Mulholland, 2007; Rocher-Ros et al., 2020). Only the lower reach of the small stream, which had small metabolic fluxes, showed a minimal change in GPP between years and lower |ER| in the dry year. These results suggest that delayed transitions to baseflow in wetter years can sharply curtail the duration and magnitude of stream metabolic activity across diverse stream conditions.

4.2. Using Metabolism to Quantify Stream Ecosystem Function in Turbulent Streams

Stream metabolism can be challenging to quantify in streams with low signal quality (e.g., high K_{600} and low GPP; Appling et al., 2018; Ulseth et al., 2019; Battin et al., 2023). We obtained reasonable metabolism estimates for both GPP and ER from 62% of days in the large stream and just 37% of days in the small stream (Table S5 in Supporting Information S1). Daily GPP and ER estimates were larger and more strongly correlated in the large stream relative to the small stream, indicating that autotrophic production may be a more important component of overall stream metabolism in the large stream. Both reaches of the small stream were more shaded from sunlight by canopy cover and topographic features, reducing light availability for autotrophic production and making estimating metabolism more challenging. Even on days with detectable GPP and ER, we found very weak correlations between GPP and ER, which suggests that ER was more decoupled from GPP in the small stream relative to the large stream. High gas exchange rates and hydrologic variability in the small stream dampened diel dissolved O_2 signals, making GPP difficult to detect, especially in the lower reach. Consistent with this limitation, the upper reach of the small stream exhibited only a modest productivity signal, with 16 days exceeding 0.5 g O_2 m^{-2} d^{-1}. Therefore, these metabolism estimates may underrepresent annual ER in the small stream by missing times of potentially higher ER, because we were not able to estimate metabolism during periods of high flow or strong hydrologic variability, when GPP is low and oxygen dynamics are dominated by gas exchange and turbulence (Hotchkiss et al., 2015; Lupon et al., 2019; Pathak & Demars, 2023; Rocher-Ros et al., 2020).

4.3. Mountain Stream Influence on Downstream Nitrogen Export

Contrary to expectations based on watershed size and discharge, the small stream had consistently higher inorganic nitrogen supply relative to the large stream (Figure 5). Past work has demonstrated that even small streams can exert a large influence on nitrogen dynamics by retaining and transforming up to half of the terrestrial organic and inorganic nitrogen inputs to these streams (Alexander et al., 2002; Bernot et al., 2010; Peterson et al., 2001). Nitrogen concentrations were generally low in both pore and surface waters across streams (Table 1); however, the small stream had higher mean nitrate and ammonium concentrations in sediment and water samples and exported >85% more inorganic nitrogen downstream than the large stream despite comparable and minimal urban land use in the watersheds (Figures 5a–5h). Nutrient supply estimates were derived from WRTDS-infilled nitrate and ammonium concentrations (Section S1.3 in Supporting Information S1) and therefore contained additional uncertainty. Infilled nitrogen concentration estimates were generally less variable than observed concentrations (Figure S2 and Table S2 in Supporting Information S1), underestimating the high concentrations and overestimating low concentrations. Therefore, the nitrogen dynamics reported here are likely more reflective of the central tendencies of the streams than fully descriptive of either high or low extremes.

Peak GPP and concomitant nitrogen demand occurred in fall and summer in the large and small streams, respectively, whereas peak nitrate or ammonium supply typically occurred about a month earlier in each stream (Figure 4). This mismatch, combined with heightened water yield and flow, may explain the pronounced increases in nitrogen supply from dry to wet water years (Figure 5). We expect that the prolonged high flows inhibited algal persistence and delivered excess nitrogen downstream. This is supported by our observation of a large increase in both ammonium and nitrate supply alongside a substantial decrease in the proportion of nitrogen retained between the upper and lower reaches in the large stream. We observed an even greater increase downstream in nitrate and ammonium supply in the small stream, which continued to exhibit low nitrogen retention between the upper and lower reaches. The shift in inorganic nitrogen supply and retention from dry to wet conditions indicates a decoupling between the timing of peak nitrogen availability and peak biological demand (Battin et al., 2016). This decoupling highlights the mechanistic limitations of headwater streams in

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retaining nutrients, especially during high-flow periods when physical transport exceeds biological assimilation (Day and Hall, 2017).

Nitrogen uptake and processing is expected to increase with increases in autotrophic production and algal biomass (Hall & Tank, 2003; Hanrahan et al., 2018; Heffernan & Cohen, 2010; Newbold et al., 2005; Tank et al., 2018); however, the effect of autotrophic production on nitrogen removal is context-dependent and can be modulated by light availability (Dodds et al., 2000; Vincent et al., 2025). Although assimilatory uptake has the potential to exceed nitrogen supply under favorable conditions (Dodds et al., 2002; Hanrahan et al., 2018; Ribot et al., 2013), this only occurred in our study during short windows when streamflow was low and light availability supported high GPP. During wet years, when flows were elevated and light availability was constrained, hydrologic transport overwhelmed in-stream assimilation, especially in the small streams. These physical changes coincided with a decline in sediment OM across most reaches (Figure S14 in Supporting Information S1), indicating reduced microbial habitat during high flows, which likely further limited biological nitrogen demand and contributed to low ammonium retention. Together, these dynamics provide evidence for how hydroclimatic conditions govern the timing and magnitude of nitrogen processing and reinforce the contrasting roles of small, nitrogen-exporting streams and larger, nitrogen-retaining streams in mediating watershed-scale nutrient dynamics (Figure 6).

5. Conclusion

Nutrient export in streams reflects the balance between nutrient deposition and availability to nutrient retention and assimilation, based on supply and biological demand over time (Bernhardt et al., 2005; Brookshire et al., 2009; Seybold & McGlynn, 2018). However, hydroclimatic disruptions to streamflow through reduced snowpack, increased evapotranspiration, earlier peak flows, and longer, more severe droughts are expected to disrupt biogeochemical cycling in freshwater ecosystems (Carroll et al., 2024; Dettinger et al., 2015; Patterson et al., 2022; Siirila-Woodburn et al., 2021). Many watersheds historically characterized by winter freezes are now experiencing mid-winter snowmelt, winter rainfall, and winter runoff or flooding (Hayhoe et al., 2018; Hock et al., 2019; Musselman et al., 2021) that could affect carbon and nitrogen cycling within streams by enhancing nitrogen mobilization during periods of low biological demand (Coble et al., 2019; Seybold et al., 2022; Winter et al., 2022).

Our study highlighted the ecological consequences of hydroclimatic whiplash, in which abrupt shifts between drought and extreme wet conditions restructured stream metabolism and nitrogen demand (Figure 6). During dry years, lower flows and increased light availability amplified productivity and nitrogen retention, particularly in the larger, more light-exposed stream. In contrast, prolonged high flows during the wet year shortened the growing season and suppressed biological demand, creating a phenological mismatch that limited nitrogen assimilation. As a result, the large stream shifted from a nitrogen sink during dry years to reduced retention during wet conditions, while the small stream remained a persistent nitrogen source, particularly for ammonium. These findings demonstrate that the capacity of streams to retain nutrients can be strongly influenced by hydroclimatic volatility, with important implications for predicting watershed-scale nitrogen export under changing climate regimes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The code for project analysis is available on Zendo (Loria, 2026) and on GitHub (https://github.com/kellyloria/Tahoe-streamflow-and-nearshore-metabolism). Data published at the Environmental Data Initiative (Loria et al., 2026).

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